

Fig. 1: Expected growth rate of the Indian 3D printing industry in FY 2016-17

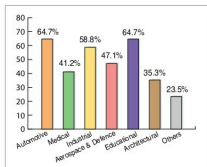


Fig. 2: Rating of applications sectors that drive the growth of 3D printing in India

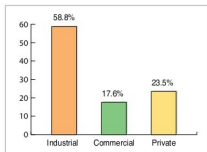


Fig. 3: Target customer groups for 3D printing industry in India

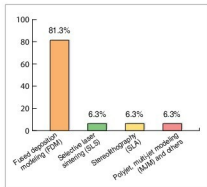


Fig. 4: 3D printing technology usage pattern in India

Currently, the Indian market is maturing. Various industries have already started using or are ready to

| EMERGING USAGE OF 3D PRINTING           |                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sector                                  | Emerging applications                                                                                                                                                                                                                                                                                                                                            |
| Automotive and industrial manufacturing | <ul style="list-style-type: none"> <li>Consolidate many components into a single complex part</li> <li>Create production tooling</li> <li>Produce spare parts and components</li> <li>Faster product development cycle with rapid prototyping, form and fit testing</li> </ul>                                                                                   |
| Aerospace                               | <ul style="list-style-type: none"> <li>Create complex geometry parts not possible with traditional manufacturing</li> <li>Control density, stiffness and other material properties of a part; also grade such properties over a part</li> <li>Create lighter parts</li> </ul>                                                                                    |
| Pharma/ Healthcare                      | <ul style="list-style-type: none"> <li>Plan surgery using precise anatomical models based on CT scan or MRI</li> <li>Develop custom or thopaeic implants and prosthetics</li> <li>Use 3D printed cadavers for medical training</li> <li>Bio-print live tissues for testing during drug development</li> </ul>                                                    |
| Retail                                  | <ul style="list-style-type: none"> <li>Create custom toys, jewellery, games, home decorations and other products</li> <li>Print spare or replacement parts for auto or home repair, for example</li> </ul>                                                                                                                                                       |
| Sports                                  | <ul style="list-style-type: none"> <li>Create complex geometry and shape not possible with traditional manufacturing</li> <li>Create custom protective gear for better fit and safety</li> <li>Create custom spike plates for soccer shoes based on biomechanical data</li> <li>Create multi-colour and multi-material prototypes for product testing</li> </ul> |

(Source: PwC Technology Forecast)

use this technology, as they understand the benefits associated with 3D printing. Though the 3D printing service industry is growing at a good pace, the 3D printer manufacturing sector will require a push. Compared to the global 3D printer manufacturers, the Indian players have yet to establish their brands and lack volumes.

However, in the past few years, a significant number of companies in India have been coming forward to explore this segment. Around two to three years ago, there were very few companies in this industry, but now quite a few companies are manufacturing 3D printers. People are getting easy access to 3D printing and the market's expectations are very high.

## Growth drivers

In the past year, automotive applications have accounted for the highest revenue share in the Indian 3D printer market. According to survey participants, during FY 2016-17, automotive and educational sectors will also emerge as major growth drivers (Fig. 2) along with industrial sector. Besides these, architectural, aerospace, medical and defence related

applications will also see growth. Other niche applications include arts and crafts, interior decoration, fashion accessories, footwear designs, jewellery designs, animation and gaming, furniture and modelling.

Demand for 3D printers in India is primarily driven by Tier-I cities, which account for major potential growth for domestic manufacturers, local assemblers and distributors. The majority of survey participants targeted industrial customers in these areas (Fig. 3). However, private and individual customers are also getting interested in 3D printers.

At present, India primarily imports 3D printers from countries such as China, the USA and Germany. However, with government initiatives to boost domestic manufacturing, many more local players are expected to emerge soon.

The rapidly growing user base of professional CAD/CAM design software like Autodesk, Solidworks and Rhino allows engineering designs to be made in the form of software models that can be directly used as inputs by the CNC and 3D printing industry. The adaptation of digital CAD processes in many of the traditional manufacturing